IEEE Open Journal of
Power Electronics

Received 10 November 2025; accepted 4 December 2025. Date of publication 10 December 2025;
date of current version 24 December 2025. The review of this article was arranged by Associate Editor Peng Fang.

Digital Object Identifier 10.1109/OJPEL.2025.3642554

An Interleaved Current-Fed PFC Flyback Converter With Voltage Follower Characteristic

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This work was supported in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES), Brazil, under Finance Code 001, and in part by the National Council for Scientific and Technological Development (CNPq).

ABSTRACT This article proposes an interleaved modified flyback converter employing pulse frequency modulation (PFM) for power factor correction (PFC) applications. The topology integrates a switching cell (SC) and coupled inductors, enabling soft commutation over a wide frequency range, a current-fed configuration, and low input current ripple. The reverse recovery losses are eliminated as the diodes block naturally, and the switches operate under zero-voltage transition (ZVT). The converter exhibits resistive input behavior and functions as a voltage follower, allowing for simplified control with a reduced component count and a single voltage feedback loop. The main features of the proposed converter include high voltage gain, galvanic isolation, and efficient AC-DC power conversion with improved power quality. A detailed analysis of the operating principle, mathematical modeling, and design procedure is presented. Experimental validation uses a 350 W prototype operating at 69 kHz with a 127 V RMS input and a 400 V output. The results demonstrate a peak efficiency of 95.1%, a low output voltage ripple, a near-unity power factor (PF), minimal zero-crossing distortion, and a low total harmonic distortion (THD).

INDEX TERMS Interleaved converter, isolated converter, power factor correction (PFC), pulse frequency modulation (PFM), rectifier.

I. INTRODUCTION

With the rise of electrification and increased use of DC loads, AC-DC converters have become essential to connect the AC grid to DC powered devices for adapters and chargers in both residential and industrial applications, including computers, smartphones, LED drivers, data centers, telecommunication systems, motor drivers, battery chargers, UPS systems, electric mobility, battery energy storage for distributed generation, etc. [1], [2], [3], [4], [5], [6], [7]

The number of power conversion devices connected to the grid has recently increased, driving the need for power electronic devices that offer high efficiency, high power density, and low cost [6], [7], [8], [9], [10], [11]. However, the growing demand for electrical energy can impose challenges to the AC power grid if not correctly managed, leading to issues such as voltage distortion, power losses, low power factor (PF), network overloading, electromagnetic interference (EMI), and degradation of generators, transformers, and capacitors. These complications can also interfere with the performance of other electronic equipment [12], [13], [14], [15].

To address the power grid challenges, all converters with a power rating greater than 75 W must comply with the International Electrotechnical Commission IEC61000-3-2 standard, which limits harmonic distortion in the power system. As a result, it is essential to develop converter topologies with power factor correction (PFC), controlled power factor (PF), and low total harmonic distortion (THD) [3], [4], [7], [9], [11], [12], [13], [14], [15], [16], [17], [18]. Although passive filters can meet these standards, active PFC techniques, which utilize switching devices, have become the focus of research due to advances in power semiconductor technology and microelectronics [19], [20].

Single-phase AC–DC converters can be classified into two categories based on the number of power processing stages: two-stage and single-stage converters. The first stage performs rectification and power factor correction (PFC) in two-stage

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topologies, converting the AC input into a constant DC voltage link. In contrast, the second stage is a DC-DC converter that regulates the output voltage. A large capacitor is required between the two stages [1], [7], [8], [11], [13], [19], [20], [21], [22], [23]. These converters offer high performance, including nearly unit PF, low harmonic distortion, and regulated output voltage. However, they come with drawbacks such as increased component count, high cost, low efficiency, and high circuit complexity, as each stage requires independent control design [1], [7], [8], [10], [11], [13], [18], [19], [20], [21], [22], [23].

Single-stage PFC converters have been proposed to simplify the design and address the limitations of two-stage systems by combining power factor correction (PFC) and voltage regulation into a single, simplified stage. By eliminating one stage, these converters reduce the number of components, simplify the control system, lower costs and volume, and improve efficiency [1], [8], [10], [17], [18], [19], [20], [21], [22], [23], [24], [25].

The flyback converter is the most popular single-stage topology for galvanic isolation and PFC applications due to its simple structure, step-up and step-down capability, adjustable output gain, small size, low component count, low cost, and ease of control [1], [5], [25], [26], [27], [28]. However, it faces challenges such as hard switching (which increases losses), high voltage spikes across switches due to transformer leakage inductance, and the need for a large output capacitor to minimize the voltage ripple. As a result, it is typically used in low-power applications [2], [15], [21], [26], [28].

If the voltage spikes across the switches are not limited to a safe level, they can cause damage to the components. To mitigate this issue, active clamp or snubber circuits are commonly used. Among the various solutions proposed, the dissipative resistor-capacitor-diode (RCD) snubber is the simplest and most traditional method [7], [11], [18], [21].

References [27], [30] and [31] present topologies similar to the proposed converter. In [27], an interleaved flyback converter with two parallel-connected converters is proposed. It requires an auxiliary inductor at the output for zero-voltage switching (ZVS) and diode reverse recovery mitigation, suffers from high voltage spikes across the switches, and has low efficiency. The push-pull current-fed converter in [30] employs an isolation transformer with PWM-driven switches. It requires auxiliary commutation inductors for ZVS, introduces additional topological states, presents diode reverse recovery losses, and lacks voltage follower characteristics. For PFC applications, it also demands an extra current control loop. The half-bridge converter [31] features an isolation transformer and PWM switches. However, it requires voltage and current feedback loops, additional capacitors, and an output inductive filter, while the switches operate with different reference potentials.

A single-stage bridgeless converter is proposed in [18], offering high PF, low voltage spikes across the switch, and a low component count. However, it requires a high input filter inductance, large capacitances in the snubber circuit and at the output load, and relatively low efficiency. [10] proposes a single-stage flyback converter with an active clamp for step-down applications. The results indicate that ZVS is achieved, with no spikes over the switch. However, it requires four switches, presents low efficiency, and has relatively low PF.

This paper proposes a novel interleaved flyback-based AC-DC converter with pulse frequency modulation (PFM) and PFC characteristics, referred to as (IFC-PFM PFC). The proposed converter is current-fed and features self-balancing of the current at the interleaved stages, eliminating the need for any current or voltage control loop. It also exhibits a natural voltage follower characteristic, making it possible to develop a control system that only measures the generated output voltage, thereby reducing the costs and effort of data processing. The proposed converter has several other significant advantages, as follows.

1) The input current ripple is twice the switching frequency, and its amplitude diminishes as the power decreases, being inversely proportional to the switching frequency.
2) The input current is split between the switches.
3) It supports both series and parallel output connections.
4) It maintains a low turn ratio.
5) It achieves zero voltage transition (ZVT).
6) It has a low component count.

Compared to the conventional interleaved flyback converter, the proposed topology eliminates the need for an input filter capacitor. Additionally, it presents lower voltage stress across the switches, along with soft switching, which reduces losses and EMI. The current-fed structure provides continuous input current, allowing for natural current sharing between the interleaved phases. The secondary windings' series connection permits greater voltage gain without excessive duty cycles.

Compared to the prior research in [32], this article provides a deeper analysis and additional experimental results. The updated power level implies changes in the parameters of the coupled inductors. The theoretical losses were calculated, and the harmonic content compared with the IEC standard. The switch and diode waveforms, efficiency, THDi, and PF across the entire power range are presented, along with the control system's performance during step load responses. The comparison table has been updated.

The remainder of this paper is organized as follows. The topology is introduced in Section II. The evaluation of the principle of operation and the equations is explained in Section III. The design considerations, including the snubber circuit and the analysis of the principal components, are detailed in Section IV. Section V presents the experimental results obtained from the laboratory prototype. Finally, Section VI concludes the paper.

II. PROPOSED TOPOLOGY

The primary side of the proposed converter is based on the switching cell (SC) introduced in [29], which explains the topological variations and the operation principles, and

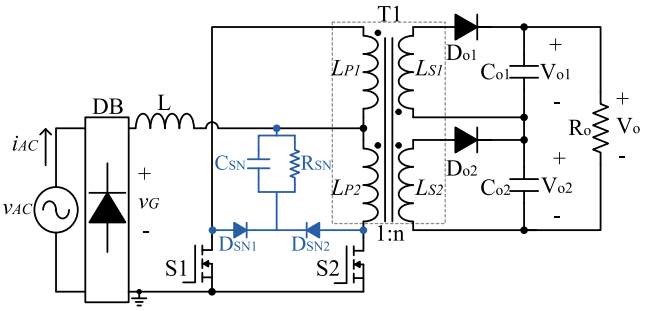


FIGURE 1. Topology of the IFC-PFM PFC.

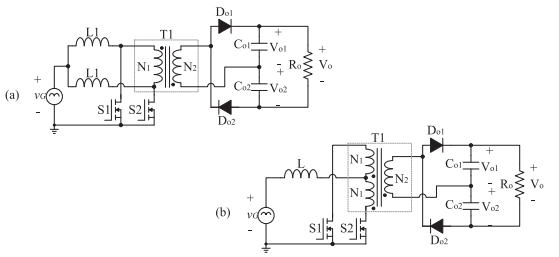


FIGURE 2. Topology variations of the proposed converter.

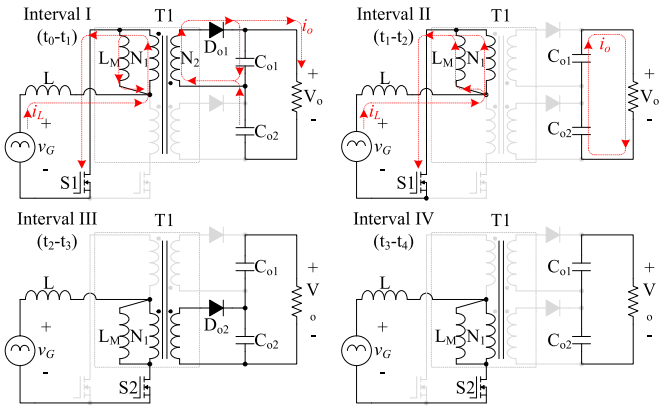


FIGURE 3. Topological states of the proposed converter.

provides a detailed mathematical analysis. The topology is introduced in Fig. 1. Additionally, no inductive-capacitive filter is required on the grid side or after the diode bridge.

The secondary side of the proposed converter is based on an interleaved flyback converter, featuring two windings connected to diodes and an output capacitor that reduces voltage ripple at the load. The adaptation of the proposed converter lies in its use of only two switches, regardless of the number of interleaved flyback cells, with control provided by Pulse Frequency Modulation (PFM), a characteristic of the applied switching cell (SC).

The power stage includes the input AC voltage, v_{AC}, the rectifying diode bridge, DB, and the rectified input voltage, v_G. L is the input inductor, operating in continuous conduction mode (CCM). $S1$ and $S2$ are switches, operating at a fixed duty cycle of 0.5 with a 180-degree phase shift, and are controlled by PFM. D_{SN1} and D_{SN2} are the snubber circuit diodes, while C_{SN} and R_{SN} are the capacitors and the resistors, respectively, designed to protect the switches.

$T1$ represents four coupled inductors mounted on a single core, consisting of two primary windings (L_{P1}, L_{P2}) and two secondary windings (L_{S1}, L_{S2}). The magnetizing inductance seen from each primary winding is denoted as L_M, which is used in the topological state description. L_{MAG} refers to the magnetizing inductance seen from the two primary windings in series, with a total inductance of $L_{MAG} = 4L_M$. It is important to note that the magnetizing inductance enables both soft switching and voltage follower characteristics. The turns ratio, n, defined as $n = L_{S1}/L_{P1}$, is equivalent to one.

D_{o1} and D_{o2} represent the output diodes, each connected to individual capacitors C_{o1} and C_{o2}, respectively, generating the output voltage across each flyback cell, denoted as V_{o1} and V_{o2}, respectively. The output load, R_o, receives the converter's total output voltage, V_o.

Variations of the proposed topology are shown in Fig. 2. Fig. 2(a) uses model A from the SC proposed in [29], featuring two separated inductors at the input, which requires an additional core. Fig. 2(b) is based on the same SC but uses a single winding on the coupled inductor's secondary. Both variations have advantages and disadvantages, but their operation and results are identical. The proposed converter and its topological variations can be designed with parallel or series output connections.

III. PRINCIPLE OF OPERATION

Based on the exposed information, the principle of operation is analyzed following several assumptions:

1) All semiconductors are considered ideal
2) V_o remains constant during a switching cycle
3) Δi_L is negligible during a switching cycle
4) The capacitors are sufficiently large to maintain a constant voltage during one switching cycle.
5) $N1$ and $N2$ are identical, and $V_{o1} = V_{o2}$; thus $V_o = V_{o1} + V_{o2}$

Fig. 3 illustrates the four topological states of the proposed PFC rectifier. The converter operates symmetrically, with intervals I and II equivalent to intervals III and IV.

A diagram showing the essential waveforms of the components is presented in Fig. 4, specifying their respective levels. The intervals are defined within a switching period, supporting the development of the equations. The switching frequency and period are f_s and T_s, respectively.

Interval I (t_0–t_1)*:* At instant t_0, switch $S1$ is turned on, $S2$ is turned off, and L_M starts discharging. Diode D_{o1} begins to conduct according to the expressions detailed in Fig. 4, while D_{o2} remains in its blocked state. $S1$ rapidly reaches its peak current level, equal to the input current. The voltage across D_{o2} equals the output voltage.

At the end of this interval, the current through diode D_{o1} becomes zero and blocks with zero current switching (ZCS), avoiding reverse recovery loss, as highlighted in Fig. 4.

Interval II (t_1–t_2)*:* Switch $S1$ maintains a constant current, while $S2$ and the diodes remain OFF. Since $L >> L_M$, the

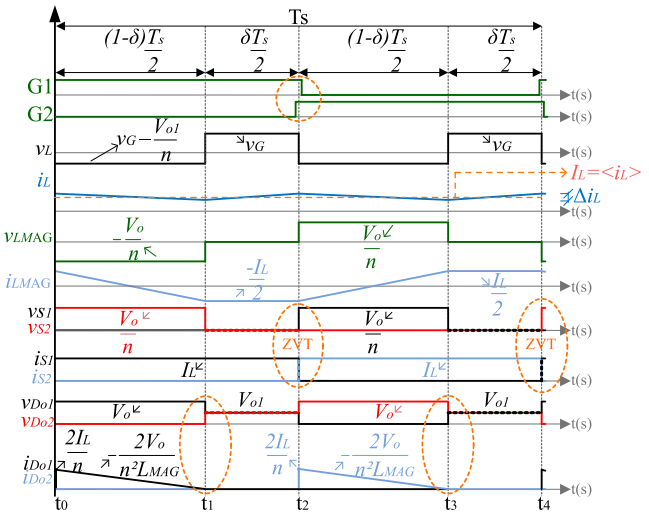


FIGURE 4. Steady-state waveforms of the proposed converter. The arrows indicate the more representative values and slopes.

voltage across the input inductor, L, equals the input voltage, making the voltages across the primary of $T1$ and L_M negligible. Applying Kirchhoff's law, the voltage at $S2$ is also zero.

During this interval, L_M stores energy, while the magnetizing inductance maintains a constant current equal to half the input current due to the steady voltage applied. Meanwhile, the capacitors supply energy to the load, which is now connected in series. The voltage across the diodes is half of the output voltage, while the secondary windings of $T1$ also experience zero voltage.

Considering the explanation above, the remaining intervals are similar, with the conducting components alternating between $S2$ and D_{o2} in intervals III and IV. The topology requires that, at any moment, at least one switch is closed, which is why the gate signals overlap.

A. STEADY STATE ANALYSIS

A steady-state operation is assumed for the analysis of the converter. The leakage inductance, L_{LK}, and the snubber circuit are neglected for simplification.

Applying the principle of voltage balance to the input inductor L, and considering its waveforms in Fig. 4, (1) is derived. The relationship between the individual output voltages and the rectified input voltage is expressed in (2), while the effective duty, δ, is defined in (3).

$$\left(v_G - \frac{V_{01}}{n}\right)(1-\delta) + v_G \cdot \delta = 0. \tag{1}$$

$$V_{o1} = V_{o2} = \frac{n \cdot v_G}{1-\delta} \tag{2}$$

$$\delta = \frac{V_o - 2n \cdot v_G}{V_o}. \tag{3}$$

The effective duty cycle corresponds to the time ratio during which the magnetizing inductance stores energy within a switching cycle, as illustrated in Fig. 3.

B. INPUT CURRENT ANALYSIS

Fig. 3 highlights two crucial points in the magnetizing inductance current waveform, L_{MAG}, which are reintroduced in (4).

$$\begin{cases} I_{L_{MAG}}(t_0) = \frac{I_L}{2} \\ I_{L_{MAG}}(t_1) = -\frac{I_L}{2} \end{cases} \tag{4}$$

During interval I, the voltage across L_{MAG} corresponds to V_o/n, resulting in (5).

$$I_{L_{MAG}}(t_1) = I_{L_{MAG}}(t_0) - \frac{1}{L_{MAG}} \int_{t_0}^{t_1} \frac{V_o}{n}\, dt \tag{5}$$

Rearranging the variables, δ is obtained in (6).

$$\delta = 1 - \frac{2n \cdot L_{MAG} \cdot I_L}{V_o \cdot T_s}. \tag{6}$$

Using (3) and (6), the average input inductor current is derived in (7).

$$I_L = \frac{v_G \cdot T_s}{L_{MAG}} = \frac{v_G}{L_{MAG} \cdot f_s} = \frac{V_P |\sin(\omega t)|}{L_{MAG} \cdot f_s} \tag{7}$$

where V_P is the peak input source voltage. According to (7), the input current depends only on the input voltage, the magnetizing inductance of $T1$, and the switching frequency, confirming that the converter exhibits a voltage follower behavior. The equivalent input resistance is $R_e = L_{MAG} f_s$.

While these equations describe the ideal behavior of the input current, deviations occur due to intrinsic semiconductor capacitances and component dynamic inaccuracies.

C. NON-IDEALITY OF COMPONENTS

Real semiconductors exhibit intrinsic capacitances, resulting in deviations from ideal operation. In the proposed converter, these capacitances alter the effective duty cycle, impact the PF, and affect the linearity between the switching frequency and the output power. The affected time intervals occur when switches turn off and the diodes block.

When the diode blocks, the primary winding current is entirely absorbed by L_{MAG}, which is in parallel with the intrinsic capacitances of the diode, C_D, and the switch, C_S. The energy stored in C_D and C_S is transferred to L_{MAG}, generating a freewheeling current, I_{FW}. The duration of the first time interval is determined by (8), while the freewheeling current, by (9).

$$t_{i1} = \pi \sqrt{L_M (C_S + C_D)} \tag{8}$$

$$I_{fw} = \frac{V_o}{2} \sqrt{\frac{(C_S + C_D)}{L_M}} \tag{9}$$

The freewheeling current is directly proportional to the semiconductor capacitances and inversely proportional to the converter's power level. A second time interval occurs when the switch turns off, during which the input and freewheeling currents drain the stored energy, transferring it to the diode.

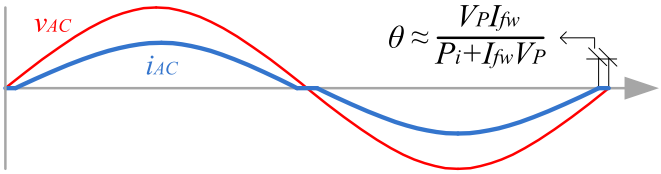


FIGURE 5. Angular interval where current remains null, before and after the zero crossing.

This interval is determined by (10).

$$t_{i2} = \frac{V_o(C_S + C_D)}{I_L + I_{fw}} \tag{10}$$

The combined time intervals reduce the effective switching period, T_e, as expressed in (11). For simplicity, t_{i2} is neglected since its impact is minimal.

$$f_e = \frac{1}{T_e} = \frac{1}{T_s - t_{i1}} \tag{11}$$

Consequently, the input current is rewritten in (12).

$$I_L = \begin{cases} \frac{v_G}{L_{\mathrm{MAG}} \cdot f_e} - 2I_{\mathrm{fw}}, & \frac{v_G}{L_{\mathrm{MAG}} \cdot f_e} > 2I_{\mathrm{fw}} \\ 0, & \frac{v_G}{L_{\mathrm{MAG}} \cdot f_e} \leq 2I_{\mathrm{fw}} \end{cases} \tag{12}$$

When the first term is smaller than the freewheeling current, the input inductor does not conduct. The angular interval during which the current remains zero is determined in (13). Rearranging the variables leads to (14).

$$\frac{V_P |\sin(\theta)|}{L_{MAG} \cdot f_e} = 2I_{fw} \tag{13}$$

$$\sin(\theta) = \frac{L_{MAG}.f_e \cdot 2 \cdot I_{fw}}{V_P} = \frac{V_P \cdot I_{fw}}{P_i + V_P \cdot I_{fw}} \tag{14}$$

Finally, the approximate expression for θ is derived in (15), and illustrated in Fig. 5. The measurement is in radians.

$$\theta = arcsin\left(\frac{V_P \cdot I_{fw}}{P_i + V_P \cdot I_{fw}}\right) \approx \frac{V_P \cdot I_{fw}}{P_i + V_P \cdot I_{fw}} \tag{15}$$

This distortion impacts the converter's THD and PF. According to (15), θ is inversely proportional to the input power. As operating power decreases and the switching frequency increases, the angular interval becomes larger, leading to a lower PF and higher THD.

IV. CONTROL STRATEGY

According to (7) and (12), the proposed converter exhibits an input voltage follower characteristic, where the switching frequency determines the input current and power.

The proposed converter requires only a single output voltage feedback loop, as detailed in Fig. 6. The measured output voltage is compared to a reference, generating an error signal that a PI controller processes to adjust the switching period. The digital signal processor (DSP) dynamically modifies the switching frequency, maintaining a fixed duty cycle of 0.5 and a 180° phase shift between the gate signals of the switches.

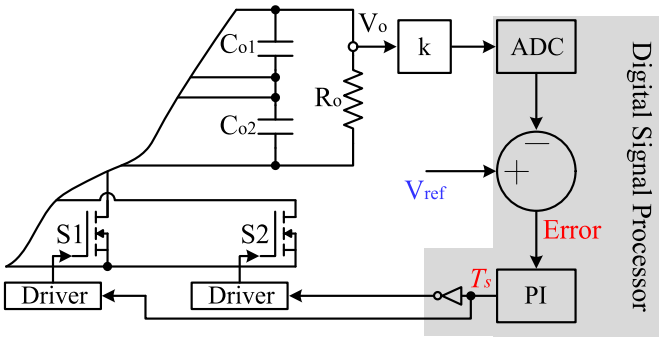


FIGURE 6. Detailed control scheme of IBC-PFM PFC.

The calculations are conducted in terms of the period, because it has a linear response to power variations.

The small-signal model of the converter is derived by analyzing power distribution across passive output components, considering minor disturbances under nominal conditions, as expressed in (16).

$$G_v(s) = \frac{\tilde{v}_o(s)}{\tilde{t}_s(s)} = \frac{V_{i_{RMS}}^2 \cdot R_o}{2L_{MAG} \cdot V_o} \frac{1}{s\left(\frac{C_o \cdot V_o}{2}\right) + 1} \tag{16}$$

where $\tilde{v}_o$ is the small signal of the output voltage, $\tilde{t}_s$ is the small signal of the switching period, $V_{i_{RMS}}$ is the RMS input voltage, C_o is the output capacitance, and R_o is the nominal load resistance. This expression is the basis for the proportional–integral (PI) controller design.

A multiphase interleaved boost converter employing the same switching cell concept was proposed in [33]. In that design, a simple PI controller combined with phase shifts between the interleaved phases was sufficient to regulate the output voltage and achieve good performance. Similarly, the proposed converter benefits from an inherently balanced structure and voltage-follower behavior, allowing for straightforward voltage-loop control without requiring current sensing. This simplicity is a practical advantage because it reduces computational effort and hardware complexity, and it also aligns with the main focus of this work, which is the converter topology and its inherent characteristics. Keeping this simple control approach shows that good performance can be achieved.

V. DESIGN CONSIDERATION AND KEY PARAMETERS

A. INPUT INDUCTOR

The inductor L is designed to reduce input current ripple. Its value can be determined using (17), which is derived from the current variation observed during the second and fourth time intervals, as well as the general voltage equation across the inductor.

$$L = \frac{\delta \cdot v_G}{2\Delta I_L \cdot f_s} = \frac{V_P |\sin(\omega t)|}{2 \cdot \Delta I_L \cdot f_s} \tag{17}$$

Where ΔI_L represents the current ripple. The inductor is built using a powder toroidal core with two parallel conductors.

B. COUPLED INDUCTORS

The coupled inductors are constructed using litz wire to eliminate the skin effect. The windings are arranged parallel, with the primaries and secondaries mounted together.

The parallel technique helps minimize leakage inductance and other unwanted parasitic magnetic effects, which must be kept as low as possible. The air gap is also carefully adjusted to achieve the desired magnetizing inductance, a key parameter influencing the input current behavior.

C. SNUBBER DESIGN

The leakage inductance resonates with the intrinsic capacitance of the switches at the switching instant, generating overvoltage spikes. These spikes are mitigated by the passive RCD snubber, as illustrated in Fig. 1.

The selected RCD snubber topology is simple but inherently dissipative, resulting in a slight reduction in efficiency. However, it was added for the safety of the switches and to demonstrate that it doesn't interfere with the converter's operation, proving that the proposed topology is a viable and competitive solution.

D. CAPACITORS

The capacitors are used to smooth the output voltage ripple. Two electrolytic capacitors are employed, one for each flyback cell. Their sizing is based on the voltage stress, as described by (18), which is a function of the output current.

$$C_{o1} = C_{o1} = \frac{I_o(\delta + 1)}{2 \cdot f_s \cdot \Delta V_c} \tag{18}$$

Where ΔV_c is the voltage ripple.

E. LOSSES ANALYSIS

The inductor and coupled inductors losses are classified into copper and core, $P_{L_{CU}}$, $P_{L_{core}}$, $P_{T1_{CU}}$, and $P_{T1_{core}}$, respectively, as determined in (19) and (20).

$$P_L = P_{L_{CU}} + P_{L_{core}} = I^2_{L_{RMS}} \cdot R_L + V_e \cdot P_{core} \tag{19}$$

$$\begin{aligned} P_{T1} &= P_{T1_{CU}} + P_{T1_{core}} \\ &= I^2_{N1_{RMS}} \cdot R_{PRI} + I^2_{N2_{RMS}} \cdot R_{SEC} + V_e \cdot P_{core} \end{aligned} \tag{20}$$

Where V_e is the core volume and P_{core} is the core loss per unit volume. R_L, R_{PRI}, and R_{SEC} are the conductors' resistances. $I_{L_{RMS}}$, $I_{N1_{RMS}}$, and $I_{N2_{RMS}}$ represent the RMS current through the inductor, primary, and secondary windings, respectively. Since there are two primary and two secondary windings, the copper losses of the coupled inductors must be doubled.

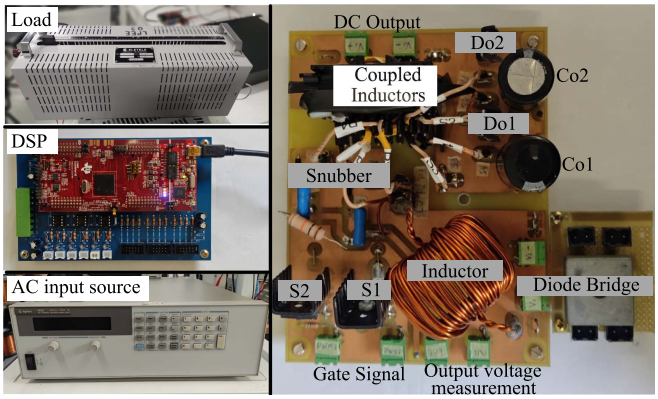


FIGURE 7. Prototype, control boards, input AC source, and load.

The switches' losses are due to conduction, switching, and capacitance, as calculated in (21).

$$\begin{aligned} P_S = I^2_{S_{RMS}} \cdot R_{DS_{on}} + \frac{1}{2} V_S \cdot I_S \cdot f_s (t_{ON} + t_{OFF}) \\ + \frac{1}{2} C_{ds} \cdot V^2_{ds} \cdot f_s \end{aligned} \tag{21}$$

Where $R_{DS_{on}}$ is its on-state resistance, t_{ON} and t_{OFF} are the turn-on and turn-off delay times, respectively, and C_{ds} is the drain to source capacitance, which can be obtained from the datasheet. I_s is the current at the switching instant, V_s is the maximum voltage stress, and V_{ds} is the drain source voltage.

The diodes' conduction losses are determined by (22).

$$P_D = I_{D_{AVG}} \cdot V_F \tag{22}$$

where V_F is the diode forward voltage and $I_{D_{AVG}}$ is the average current.

The capacitor conduction losses are given by (23).

$$P_C = I^2_{C_{RMS}} \cdot ESR \tag{23}$$

Where ESR is their equivalent series resistance, and $I_{C_{RMS}}$ is the RMS current.

Finally, the snubber losses are calculated in (24).

$$P_{SN} = \frac{1}{2} \frac{L_{LK} \cdot f_s \cdot I^2_L \cdot V_{SN}}{V_{SN} - \frac{V_o}{n}} \tag{24}$$

Where V_{SN} is the maximum voltage spike admitted over the switches, equivalent to 500 V, and L_{LK} is the measured leakage inductance of the coupled inductors.

VI. EXPERIMENTAL RESULTS

To verify the operation of the proposed converter, several tests were conducted on the 400 V/350 W prototype, depicted in Fig. 7. The AC-DC converter was designed based on the electrical parameters listed in Table 1, which also displays the characteristics of the components important for loss calculation. The waveforms were taken with a 450 Ω resistive load operating at a switching frequency of 69 kHz.

The input current and voltage waveforms are presented in Fig. 8. The zero-crossing distortion is illustrated for rated

TABLE 1. Electrical Parameters of the Prototype

Parameters	Values
Input AC voltage RMS	V_i = 127V / 60Hz
Output voltage	V_o = 400V
Output power	P_o = 350W
Switching frequency	f_s = 69kHz to 250kHz
Diode Bridge (DB)	KBPC3508; $V_{F_{DB}}$ = 0.7V
Inductor	L=500µH; 2x15AWG; 58 turns; Powder toroidal 2xCS400060
Coupled Inductors	$N1/N2$=25/25; n=1; lg=0.11mm; $L_{P1,2}$=$L_{S1,2}$; Litz1x180x0.08mm; L_M=134µH; L_{MAG}=540µH; L_{Lk}=1.5µH; ETD 44/22/15–3C90
Switches (SiC MOSFET); S_1, S_2	C2M0025120; $R_{DS_{on}}$ = 25mΩ C_{ds} = 375pF; t_{fall} = 20ns
Output capacitors; C_{o1}, C_{o2}	470µF / 250V; Electrolytic
Output Diodes; D_{o1}, D_{o2}	MUR 860; V_{F_D} = 0.75V;
Snubber Diodes; D_{SN1}, D_{SN2}	MUR 160
Snubber Capacitor; C_{SN}	5.6nF / 650V; Film
Snubber Resistor; R_{SN}	40kΩ
Drivers	2 x FOD 3180
DSP	F28379D Texas Instruments

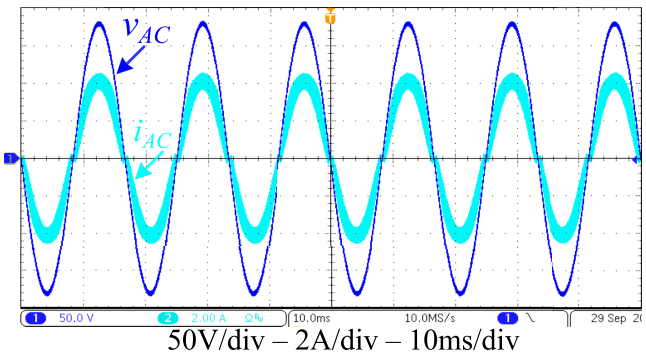


FIGURE 8. Input voltage (v_{AC}) and input current (i_{AC}) for rated power operation.

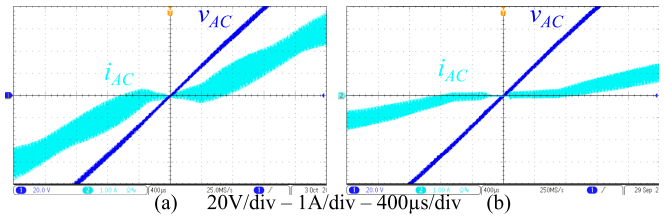


FIGURE 9. Input voltage and current during the zero crossing. (a) 100% load, f_s = 70 kHz. (b) 41% load, f_s = 133 pkHz.

power at 69 kHz in Fig. 9(a) and for 150 W at 133 kHz in Fig. 9(b). The estimated angular interval θ, during which the current remains null at zero crossing is approximately 7° (300 μs) for 350 W and 13° (586 μs) for 150 W. It increases as operating power decreases, as determined by (15) and Fig. 5.

The odd harmonic components of the current at 350 W and 150 W, compared to IEC 61000-3-2 Class A and Class D limits, are presented in Fig. 10. The proposed converter complies with the regulations by a significant margin.

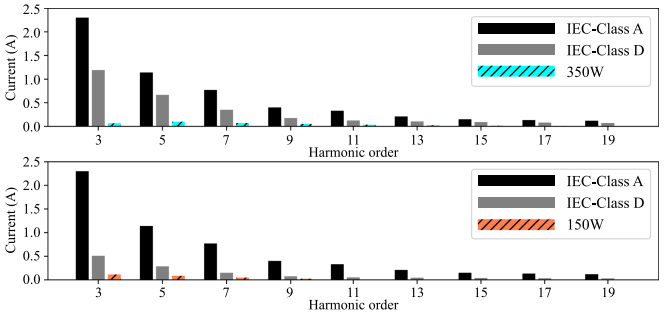


FIGURE 10. Input current harmonics for 100% and 41% of the load in comparison with IEC 31000-3-2 Class A and D.

TABLE 2. Losses and Efficiency Analysis

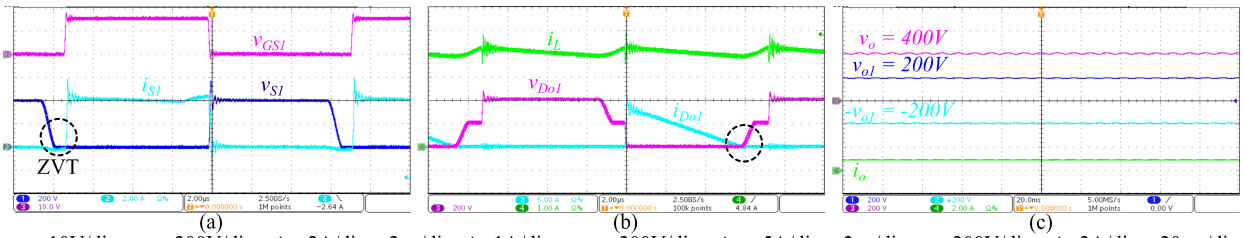
Loss calculation	Total
$P_L = P_{L_{CU}} + P_{L_{core}} = I^2_{L_{RMS}} R_L + V_e P_{core}$	1.6W
$P_{T1} = 2I^2_{LM_{RMS}} R_{PRI} + 2I^2_{N2_{RMS}} R_{SEC} + V_e P_{core}$	3.2W
$P_S = I^2_{S_{RMS}} R_{DS_{on}} + \frac{1}{2} V_S I_S f_s (t_{OFF}) + \frac{1}{2} C_{ds} V^2_{ds} f_s$	5.7W
$P_{DB} = I_{DB_{AVG}} \cdot V_{F_{DB}}$	3.4W
$P_D = I_{D_{AVG}} \cdot V_{F_D}$	1.3W
$P_C = I^2_{C_{RMS}} \cdot ESR$	0.8W
$P_{SN} = \frac{1}{2} \frac{L_{LK} \cdot f_s \cdot I_L^2 \cdot V_{SN}}{V_{SN} - \frac{V_o}{n}}$	2.2W
$P_{LOSS} = P_L + P_{T1} + P_S + P_{DB} + P_D + P_C + P_{SN}$	18.2W
$\eta = 100\% \frac{P_o}{P_o + P_{LOSS}}$	95.06%

The voltage, current, and gate signal for switch $S1$ are shown in Fig. 11(a). It starts conducting with ZVT and immediately reaches its peak value, which is equal to the input current. The voltage across the switch equals the output voltage, with a designed overvoltage spike of approximately 30% of the output voltage, which is limited by the RCD snubber.

Fig. 11(b) displays the voltage and current waveforms of the output diode D_{o1}. The reverse blocking voltage alternates between V_o and $V_o/2$, and the non-dissipative commutation is highlighted. The inductor current is also shown, demonstrating low ripple content at twice the switching frequency.

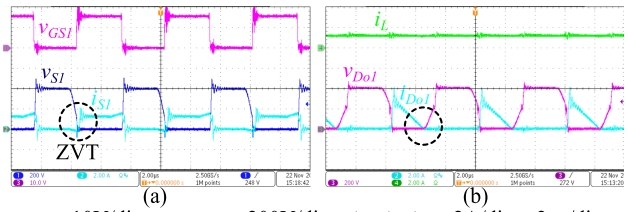
The conformity between the output stages is presented in Fig. 11(c). The voltage waveform v_{o2} is inverted and does not overlap with v_{o1} since both have the same magnitude, 200 V. The output voltage at the load maintains a mean value of 400 V, with a peak-to-peak voltage ripple of 15 V, corresponding to an oscillation of 3.75%. The output current at the load, i_o, is also illustrated.

Identical waveforms for the input inductor (i_L), switch (v_{GS1}, v_{S1}, i_{S1}), and diode (v_{Do1} and i_{Do1}), captured at a switching frequency of 160 kHz, are presented in Fig. 12. These conditions correspond to an output power of approximately 120 W. The ZVT characteristics are highlighted, and the low ripple of the input current is evident.



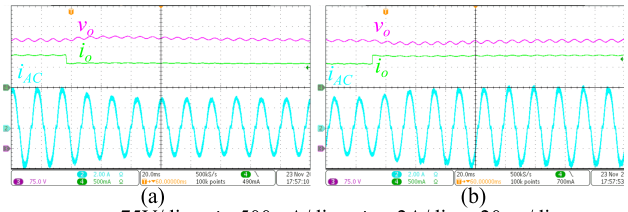
v_{GS1}=10V/div – v_{S1}=200V/div – i_{S1}=2A/div – 2μs/div i_L=1A/div – v_{Do1}=200V/div – i_{Do1}=5A/div – 2μs/div v_o=200V/div – i_o=2A/div – 20ms/div

FIGURE 11. Measurements during a switching cycle for f_s = 69 kHz and R_o = 450 Ω. (a) Voltage, current, and the gate signal at the switch $S1$; (b) Voltage and current at the diode D_{o1} and current through the inductor, i_L; (c) Voltages v_o, v_{o1}, v_{o2}, and output current, i_o.



v_{GS1}=10V/div – v_{S1}=v_{Do1}=200V/div – i_{S1}=i_L=i_{Do1}=2A/div – 2μs/div

FIGURE 12. Measured voltage, current, and the gate signal at the switch $S1$ for f_s = 160 kHz.



v_o=75V/div – i_o=500mA/div – i_{AC}=2A/div – 20ms/div

FIGURE 13. Output voltage control for step responses: (a) from 100% to 70% of the load; (b) from 70% to 100% of the load.

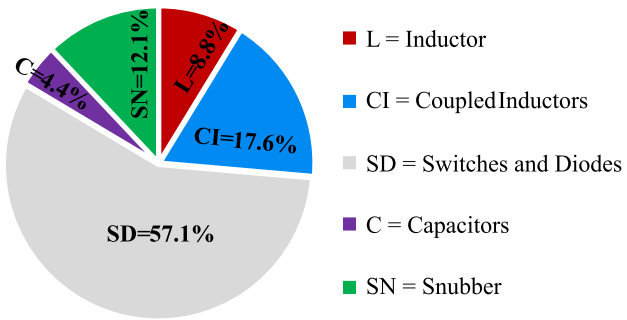


FIGURE 14. Estimated power loss distribution considering the inductor (L), coupled inductors (CI), switches and diodes (SD), capacitors (C), and the snubber (SN).

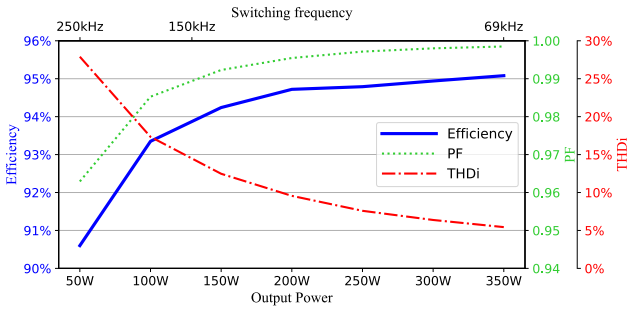


FIGURE 15. Measured efficiency, power factor, and input current THD as a function of the output power.

Fig. 13(a) presents the waveforms of output voltage, output current, and input current for a step response from 350 W to 250 W, while Fig. 13(b) illustrates the response from 250 W to 350 W. In both cases, the control system stabilizes the output voltage within a few grid cycles, presenting a limited overshoot and undershoot of approximately 14 V. The ringing characteristics resemble those in Fig. 11(c).

The component losses of the proposed converter at rated power are detailed in Table 2. The results consider the variables and parameters discussed previously, many of which are derived from component datasheets. The distribution and breakdown of losses are presented in Fig. 14, where the semiconductors account for approximately 57.1%. The RCD snubber, often a concern due to its dissipative nature, contributes nearly 12.1%.

The THDi, PF, and efficiency for different output power levels and their respective switching frequencies are shown in Fig. 15. The output voltage remains constant at 400 V, while the output power is varied. At rated power, the PF equals 0.998, the THDi equals 5.4%, and the efficiency is approximately 95.1%. As expected, THDi decreases, and PF improves at higher power levels. The converter maintains high power quality and efficiency across a wide range. It retains ZVT even at lower power levels, although switching and snubber losses become more prominent, reducing overall efficiency.

Fig. 16 illustrates the output voltage as a function of the switching frequency for three resistive load values: 400 Ω, 450 Ω, and 475 Ω. One key advantage of the proposed topology is its ability to regulate the output voltage by adjusting the switching frequency. These values were chosen around the rated operating point to highlight the converter's behavior. At lower frequencies, the output voltage and power are higher. Increasing the switching frequency, however, lowers the output voltage exponentially until it reaches a stable region, which differs for each case.

Table 3 compares the proposed converter with other isolated AC–DC PFC topologies, highlighting the number of active and passive devices, the magnetic components, the switching frequency, the rated input and output voltage and

TABLE 3. Comparison Between Different Converter Models

Reference	Number of power components				Coupled Inductor / Transformer		SF (kHz)	V_i(V) / V_o(V) / P_o(W)	Current sensor/current control loop	Voltage sensor/voltage control loop	Input current ripple frequency	THD	PF	Efficiency at maximum power
	S	D	C	L	Core	Win.								
[1]	1	8	4	1	2	4	100	110/120/120	0	1	f	5%	0.99	84%
[2]	1	10	6	4	1	2	100	220/48/100	0	1	-	2.2%	-	92.5%
[4]	3	5	5	3	1	3	100	110/48/120	-	-	-	7.6%	≈0.99	93%
[6]	6	6	5	1	1	2	20	110/200/1000	1	2	f	-	>0.99	93.4%
[7]	4	6	6	3	1	2	50	220/380/400	1	2	-	-	0.998	96%
[8]	1	6	4	2	1	3	200	126.5/40/160	-	-	f	8.51%	-	<90%
[9]	2	5	3	2	1	2	130	110/20/100	1	3	-	3.5%	-	≈94%
[16]	6	8	6	1	1	2	20	110/200/900	1	2	f	-	1	≈91%
[21]	2	6	2	1	2	4	50	110/248/200	0	1	-	1.4%	≈ 1	≈93.9%
[34]	1	10	5	0	2	5	42	120/48/100	-	-	f	-	>0.99	<90%
Proposed	2	8	2	1	1	4	69	127/400/350	0	1	2f	5.5%	0.998	95.1%

S=Switch; D=Diode; C=Capacitor; L=Inductor; Win.=Windings; SF=Switching frequency; V_i=Input voltage; V_o=Output voltage; P_o=Rated power.

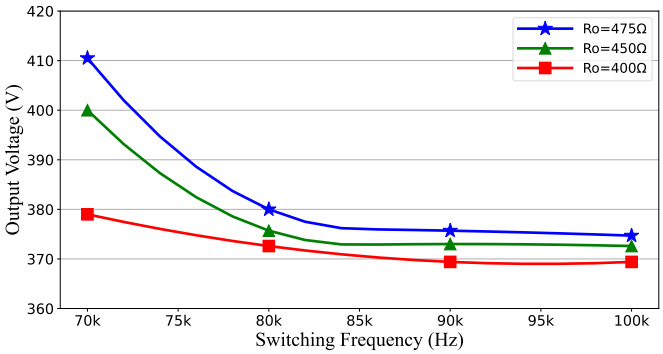


FIGURE 16. Output voltage as a function of the switching frequency for various resistive output loads.

power, the number of current and voltage sensors for control operation, input current frequency ripple, and performance indices such as THD, PF, and efficiency.

Conventional solutions typically require more switches, diodes, or inductors, which increases complexity, expense, and control load. In contrast, the proposed converter utilizes only two switches, one inductor, and one transformer to achieve isolation, high voltage gain, and competitive performance, resulting in a simpler design and fewer components.

The proposed topology requires only one voltage sensor, which simplifies digital implementation and reduces sensing losses, cost, and complexity. This contrasts with several circuits that require multiple current and voltage sensors, along with complex control loops. The proposed converter offers low input current THD (5.5%), a power factor of nearly unity (0.998), and a peak efficiency above 95%, even though it operates at a moderate switching frequency (69 kHz). These results are comparable to or better than those of topologies that operate at higher frequencies or with more devices.

This balance between simplicity, efficiency, and power quality makes the proposed solution attractive for medium-power isolated PFC applications, where cost, size, high performance, and low control complexity are important parameters.

VII. CONCLUSION

This work presented a high-voltage-gain interleaved current-fed modified flyback converter with series-connected outputs, two pairs of coupled inductors for isolation and voltage gain regulation, as well as a passive RCD snubber. The topology achieves soft switching over a wide power range, high voltage gain, low input current ripple, and natural voltage follower behavior with a reduced component count and simplified PI control that requires only one voltage sensor. Experimental validation of a 350 W prototype (127 V RMS input, 400 V output) confirmed high efficiency, near-unity power factor, low THD, and compliance with IEC harmonic limits. Compared to existing solutions, the proposed converter minimizes semiconductor and magnetic components, reduces volume and weight, and eliminates the need for auxiliary windings, especially in interleaved implementations, making it suitable for compact and efficient AC–DC PFC applications. Compared to the conventional interleaved flyback converter, the proposed design eliminates the need for an input filter capacitor, reduces voltage and current stress on the switches, and presents ZVS smooth soft commutation. The frequency modulation strategy may be seen as a limitation, although it operates at higher frequencies under lower current stress conditions.

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Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) - ROR identifier: 00x0ma614